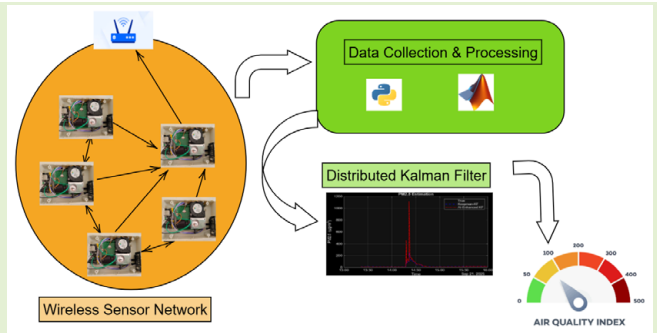


AI-Based Koopman Distributed Kalman Filter for Air Quality Monitoring Through Integrated Particulate Sensors

Kouba Chorfi¹, Jawhar Ghommam², Majdi Mansouri³, Ahlam Al-Hanai⁴, and Amani Ayeb

Abstract—This study presents an enhanced Kalman filtering (KF) approach for real-time air quality monitoring, with a particular focus on estimating inhalable particulate matter (PM_{2.5} and PM₁₀) concentrations across a distributed sensor network. The classical Kalman filter (KF) is initially employed to provide state estimation based on environmental inputs such as wind speed and humidity. While KF demonstrates effectiveness in linear systems, its accuracy degrades significantly in the presence of nonlinear pollutant behaviors, sensor noise, and dynamic disturbances that characterize real-world conditions. To address these limitations, we propose an AI-augmented Kalman filter (AI-KF) that integrates a lightweight neural network (NN) to model and compensate for the residual errors produced by the KF. This neural module operates as a data-driven corrector that enhances the filter's adaptability to nonlinear dynamics and environmental variability. Unlike the standard KF, the AI-KF can learn complex relationships between state variables and measurements, offering improved accuracy in scenarios with unmodeled nonlinearities. It demonstrates strong robustness to sensor drift and measurement noise by continuously adapting to shifting data patterns. Moreover, the AI-KF maintains the interpretability of the traditional KF framework while introducing an intelligent layer capable of real-time correction and adaptation. Experimental results on synthetic pollution data confirm that the proposed AI-KF consistently outperforms the standard KF, achieving lower root-mean-square error (RMSE) and enhanced state reconstruction. These findings highlight the potential of combining model-based estimation with neural residual learning for scalable, accurate, and robust environmental monitoring applications.

Index Terms—Connected sensors, data-driven modeling, environmental sensing, indoor air quality (IAQ) monitoring, Koopman–Kalman filter, nonlinear systems, PM_{2.5}/PM₁₀ concentration, smart sensing networks, spatiotemporal reconstruction, state estimation.



I. INTRODUCTION

INDOOR air quality (IAQ) substantially influences individuals' health and overall well-being, as they typically spend 90% of their time indoors [12], [24]. Nonetheless, IAQ and its health consequences have garnered far less focus; exposure to indoor air pollutants is more detrimental than that of outdoor air [24]. According to the World Health Organization (WHO), indoor air pollution (IAP) is attributed to the deaths of

3.8 million people annually [33]. Recent studies demonstrate that exposure to indoor air pollutants such as carbon monoxide (CO), PM_{2.5}, PM₁₀, and volatile organic compounds (VOCs) is linked to detrimental health consequences on the respiratory, neurological, and cardiovascular systems, as well as carcinogenicity and endocrine disruption [20]. Particulate matter (PM) consists of a heterogeneous mixture of solid and liquid particles that remain suspended in the atmosphere. It is commonly classified by aerodynamic diameter into PM₁₀ ($\leq 10 \mu\text{m}$), PM_{2.5} ($\leq 2.5 \mu\text{m}$), and ultrafine PM_{0.1} ($\leq 0.1 \mu\text{m}$) [21], [32], [33]. Globally, PM is recognized as a major environmental health hazard, with smaller particles posing the highest risk due to their ability to penetrate deep into the respiratory tract and even enter the bloodstream [16], [33]. Numerous toxicological and epidemiological studies have shown that short- and long-term exposure to PM is associated with a range of adverse health effects, including premature mortality, cardiovascular and respiratory diseases, and neurological disorders. Increased vulnerability has been observed among high-risk groups, such as individuals with cardiovascular disease (CVD), chronic obstructive pulmonary disease (COPD), the elderly,

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children, and pregnant women [2], [16], [20], [21], [32]. At present, although regulations have significantly enhanced outdoor air quality, indoor settings—where people primarily spend their time—remain inadequately studied and regulated. This disparity is particularly alarming in indoor settings such as elderly homes, clinics, consulting offices, and workplaces, where it is essential to uphold safe and healthy circumstances. A wide variety of pollutants can lead to IAP; therefore, it is crucial to pinpoint their main sources and concentrations, in addition to formulating methods for regulating and enhancing IAQ.

To address the limitations of existing air quality monitoring approaches, this work proposes a wireless sensor network (WSN) for indoor monitoring of PM, specifically PM_{2.5} and PM₁₀. The system deploys multiple distributed nodes across indoor spaces, enabling continuous tracking and reliable estimation of pollution levels. Multisensor fusion and tracking have a long-standing history in signal processing, control theory, and robotics, particularly when integrating data from multiple sources into a single coherent estimate. In WSNs, estimation challenges are further complicated by issues such as packet loss and communication constraints, which have been the focus of extensive recent research [28].

Prior air pollution sensor networks have targeted both large- and small-scale deployments using mobile apps, servers, and cloud platforms [11], [14], [25], [29]. These works primarily focus on data acquisition and visualization rather than robust estimation or network efficiency. For example, Devarakonda et al. [11] used Zigbee nodes to measure environmental variables and present results via a graphical interface, but did not address network optimization or advanced data fusion. This highlights the need for approaches that integrate predictive models and efficient data processing, as discussed in the following AI-based literature.

Our system adopts a mesh-based Wi-Fi network to efficiently transmit node data to a central server, reducing power consumption and architectural complexity. This builds on prior ESP-MESH work [18], which focused on hardware implementation and network stability.

From a signal-processing perspective, combining noisy measurements into robust air quality estimates requires advanced filtering. Kalman filter variants have been widely applied for this purpose: [26] used an unscented Kalman filter (UKF) to fuse measurement data with emission inventories, [27] combined a the air pollution model-chemical transport model (TAPM-CTM) with an extended fractional Kalman filter (EFKF) for inverse emission estimation and urban pollution prediction, and [13] demonstrated EKF for multisensor fusion of IAQ, integrating IAQI and humidex measurements to enhance real-time estimates and occupant comfort. This motivates our adoption of a Kalman-based approach for robust, real-time IAQ estimation.

Building on these foundations, we employ a Kalman filter to integrate sensor observations with predictions from a dynamic model. To handle nonlinearities in air quality dynamics, we extend this framework using the Koopman operator and further enhance accuracy with an AI-augmented Koopman–Kalman filter (AI-KF), where a neural residual cor-

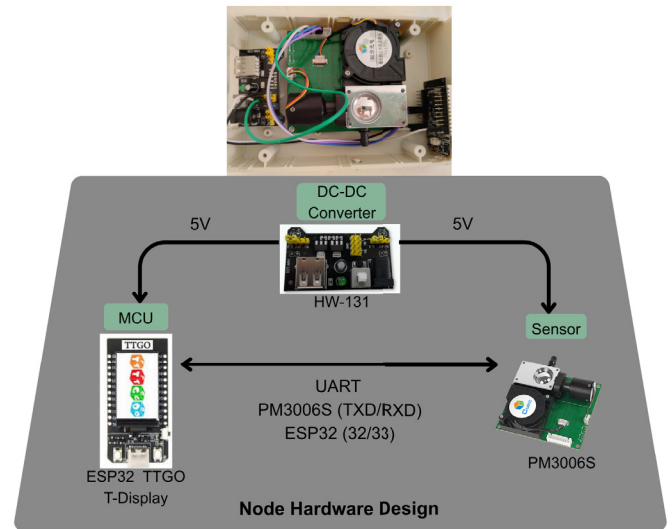


Fig. 1. Node hardware architecture integrating the PM3006S sensor, ESP32 MCU, and DC–DC power module.

rector leverages historical data. The performance of both the standard Koopman KF and AI-KF is evaluated, highlighting the benefits of AI-enhanced estimation for IAQ monitoring.

Recent studies have increasingly combined AI and IoT for IAQ monitoring and pollutant forecasting. For example, [10] demonstrated the impact of human occupancy on CO₂/VOC levels using low-cost multipollutant sensors. AiCareBreath [4] and AiCareAir [5] introduced hybrid ML/DL frameworks, including ensemble models and transfer learning, achieving high prediction accuracy across multiple pollutants. Other approaches, such as wavelet-enhanced CatBoost [3] and timezone-aware autoregressive LSTM (TAR-LSTM) [6], further improve real-time forecasting by capturing temporal dependencies and multivariate interactions.

A summary of recent AI-based air quality prediction studies from 2020 to 2024, including methodologies, pollutants measured, and performance metrics, is provided in Table I.

The remainder of this article is organized as follows. Section II details the hardware and software design of the WSN from an IoT perspective. Section III presents the theoretical methodology, including the Koopman operator, the extended dynamic mode decomposition (EDMD) for the KDKF architecture, and the implementation steps of the AI-augmented Koopman–Kalman filter. Section IV describes the system implementation, covering sensor deployment and data collection. Section V reports the results of the IAQ assessment and provides a discussion. Finally, Section VI concludes the article.

II. SYSTEM DESIGN

A. Hardware Design

To provide a clear understanding of the physical components of the system, an example of a fully assembled sensor node is shown in Fig. 1. This node integrates the sensing, processing, and power modules into a compact and functional unit, which serves as the building block for the overall network.

1) *Sensor Module*: The PM3006S particulate sensor, based on laser-scattering technology, is used for real-time

TABLE I
SUMMARY OF RECENT AI-BASED AIR QUALITY PREDICTION STUDIES (2020–2024)

Authors	Methodology	Air Pollutants Measured	R ²	RMSE
L. Zhang <i>et al.</i> [38]	Semi-supervised Bidirectional LSTM	PM2.5, PM10	–	13.6, 10.3
X. Xu & M. Yoneda [37]	LSTM-Autoencoder, Multitask	PM2.5	–	14.52
B. Liu <i>et al.</i> [22]	Spatiotemporal RNN	PM2.5	0.75	11.49
Y. Huang <i>et al.</i> [15]	PSO-BP Neural Network	AQI	0.92	1.03
H. Chen <i>et al.</i> [8]	Dual LSTM	PM2.5	0.89	14.36
N. Seong [31]	Spatiotemporal Attention Network	PM2.5	0.9	3.474
J. Chen <i>et al.</i> [9]	Adaptive Kalman Filter	AQI	–	10.37
J. Borah <i>et al.</i> [5]	Hybrid-Ensemble IoT Sensing Unit	PM2.5, PM10	0.95, 0.93	0.00949, 0.00857
J. Borah <i>et al.</i> [4]	Location-Invariant IoT Model	PM2.5, PM10	0.98, 0.99	1.61, 2.85

measurement of PM1.0, PM2.5, PM10, and total suspended particles (TSPs) in $\mu\text{g}/\text{m}^3$. Air is drawn into a sensing chamber by an internal fan, where particles crossing a linear laser beam generate scattering signals detected by a photoelectric converter. An embedded microcontroller processes these pulses and outputs factory-calibrated mass concentrations.

The module supports UART and I²C communication, with a typical response time of 1 s and a time to first reading of ≤ 8 s. It operates at 5-V DC (± 0.1 V, ripple < 50 mV), with a working current below 300 mA and a standby current below 20 mA.

The sensor pinout includes dual VCC and GND pins, UART/I²C lines (TXD/SCL and RXD/SDA), and a mode-selection pin (CTR). The CTR pin determines the communication interface; in this work, UART mode is selected for ease of integration within the multinode wireless system.

2) Real-Time Sensor Calibration: Low-cost PM sensors require robust calibration to correct for environmental interference such as humidity, sensor drift, and variable aerosol composition, challenges that static calibrations fail to address dynamically. Although the PM3006s sensor is precalibrated in the laboratory by the manufacturer, its performance degrades in real-world environments due to variable humidity, aerosol composition, and pollution sources. Our proposed AI-augmented Koopman–Kalman filter (AI-KF) directly addresses this limitation. Unlike batch-processing methods like random forest [1], [39], which require prior data cleaning, the AI-KF integrates a lightweight neural network (NN) within a Kalman filter framework. This enables real-time, dynamic calibration that continuously adapts to environmental changes and sensor drift, transforming the sensor's raw signal into a robust and accurate measurement without the need for offline preprocessing.

Remark 1: A critical step in utilizing low-cost sensor data is addressing potential data faults arising from improper calibration. While data imputation is often used to correct faulty datasets from uncalibrated sensors, our approach addresses the problem at its source. The AI-augmented Kalman filter performs real-time calibration, dynamically compensating for environmental interference and sensor drift. This eliminates the systematic errors that necessitate large-scale imputation. However, for a very small amount of data, like random signal dropouts ($< 1\%$ of data), standard linear interpolation is applied.

3) Processing and Communication Node: Each node integrates the PM3006S sensor, the power module, and

a processing and communication unit based on the ESP32 microcontroller. The LilyGo TTGO T-Display board, shown in Fig. 1, incorporates a dual-core ESP32 (up to 240 MHz) with 4 MB of flash memory, providing sufficient resources for on-node data acquisition, processing, and transmission.

The ESP32 supports Wi-Fi (IEEE 802.11) and Bluetooth (classic and LE), enabling local computation and long-range mesh networking. Its low-power architecture and integrated radios make it suitable for distributed sensing applications. The TTGO variant also includes a 1.14-in LCD for real-time visualization of sensor readings.

To ensure reliable operation of both the ESP32 and the PM3006S sensor, an independent power supply system was integrated into each node.

4) Power Supply: Each node is powered by a 9-V battery regulated through an HW-131 DC–DC converter, as shown in Fig. 1. The converter provides 3.3- or 5-V selectable outputs with a maximum current of 700 mA; in this work, the 5-V output is used to satisfy the requirements of both the PM3006S and the ESP32.

A 100-nF ceramic decoupling capacitor is placed between the sensor's VCC and GND to suppress high-frequency noise and ensure stable operation. Tests performed without this capacitor resulted in unstable readings and harmonic distortions, while proper decoupling restored normal behavior.

B. Software Design

1) System Architecture: The software adopts a layered IoT architecture: perception, network, and application layers, providing modularity, scalability, and fault tolerance.

Perception Layer: Air quality is sensed by the PM3006S and transmitted to the MCU via UART. The MCU digitizes the signals, frames them, and temporarily buffers measurements, ensuring consistent data formatting for transmission.

Network Layer: Nodes communicate via ESP-Wi-Fi-Mesh, forming a self-healing multihop network. Each node can relay packets, while the root node aggregates measurements from child nodes. This extends coverage, maintains reliability under node failures, and supports up to 1000 nodes over 15–50-m ranges.

Application Layer: The root node bridges the mesh network to a local server via HTTP. Aggregated measurements are serialized in JSON and processed with KDKF and AI-enhanced KDKF for estimation, correction, and visualization, ensuring end-to-end reliability and modularity.

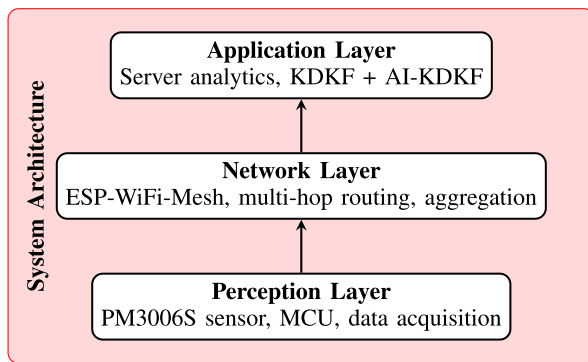


Fig. 2. Hierarchical architecture of the air quality-monitoring system, showing perception, network, and application layers.

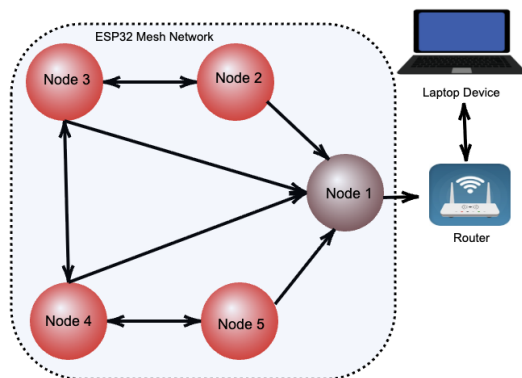


Fig. 3. Network topology.

This hierarchical architecture, Fig. 2 ensures a clear separation between data acquisition, wireless communication, and application-level services, thereby simplifying integration and guaranteeing scalability for large-scale air quality monitoring.

2) Communication Protocols:

a) *Mesh network*: ESP-Wi-Fi-Mesh organizes nodes in a self-forming, self-healing network using the AP-STA mode. Each node forwards packets to neighbors, enabling large-scale deployment without extra Wi-Fi infrastructure. The root node serves as a gateway, aggregating sensor data and transmitting it to the server over standard Wi-Fi (Fig. 3).

The server listens at a predefined or static IP, receiving data for processing with KDKF and AI-enhanced KDKF, ensuring reliable end-to-end delivery from sensors to the server.

b) *HTTP communication: root node to the server*: Aggregated measurements are serialized in JSON and sent via HTTP GET from the root node to the local server. The server parses and stores the data for further processing, maintaining continuous and reliable data delivery while preserving modularity across sensing, networking, and computation layers.

3) Data Processing:

a) *Workflow and measurements parsing*: The workflow for handling sensor measurements on each MCU is illustrated in Algorithm 1. Sensor readings received via UART from the PM3006S are first stored in local read buffers to prevent data loss during acquisition. Each UART response is transmitted in hexadecimal format within a defined frame structure as specified in the manufacturer's datasheet and contains sequential

Algorithm 1 PM Sensor Data Parsing and Processing

Input: Raw PM sensor data ($PM_{1.0}$, $PM_{2.5}$, PM_{10} , TSP) via UART

Output: Parsed PM values ready for AQI computation and transmission

Initialization:

- 1: Initialize UART communication.
- 2: Send command to open measurement mode.

Measurement Confirmation:

- 3: **while** measurement mode not confirmed **do**
- 4: Resend measurement opening command.
- 5: **end while**

Continuous Data Parsing:

- 6: **while** true **do**
- 7: Wait for new data frame.
- 8: **if** frame is valid **then**
- 9: Parse $PM_{1.0}$, $PM_{2.5}$, PM_{10} , TSP from payload.
- 10: Store values in buffers for AQI computation and network transmission.
- 11: **else**
- 12: Flag invalid frame.
- 13: **end if**
- 14: **end while**

TABLE II

AQI CATEGORIES AND HEALTH IMPLICATIONS

AQI Range	Category	Color Index
0–50	Good	Green
51–100	Moderate	Yellow
101–150	Unhealthy for Sensitive Groups	Orange
151–200	Unhealthy	Red
201–300	Very Unhealthy	Purple
301+	Hazardous	Maroon

measurements for $PM_{1.0}$, $PM_{2.5}$, PM_{10} , and TSP. Parsing is performed by iteratively extracting groups of four bytes corresponding to each particulate measurement.

Before parsing, frame integrity is verified by checking the start symbol, frame length, and command identifier. Only valid frames are processed and converted into numerical values for subsequent steps, including air quality index (AQI) calculation and aggregation for mesh transmission. This buffered parsing approach ensures robust handling of asynchronous sensor data in a multinode environment.

b) *AQI computation*: The AQI is a standardized metric developed by the U.S. Environmental Protection Agency (EPA) to communicate air quality levels [34]. In this study, AQI is computed from $PM_{2.5}$ and PM_{10} concentrations, translating measured PM levels into a scale that reflects potential health impacts.

The AQI is divided into six color-coded categories, as shown in Table II, indicating different levels of health concern and providing guidance for susceptible populations. Fine particle pollution ($PM_{2.5}$) and coarse particle pollution (PM_{10}) are significantly linked to severe health consequences, particularly influencing vulnerable groups such as the elderly, pregnant women, and children.

TABLE III

AQI BREAKPOINTS FOR PM_{2.5} AND PM₁₀ (U.S. EPA STANDARD)

Pollutant	Concentration (µg/m ³)	AQI Range
PM _{2.5}	0 – 12.0	0 – 50
	12.1 – 35.4	51 – 100
	35.5 – 55.4	101 – 150
	55.5 – 150.4	151 – 200
	150.5 – 250.4	201 – 300
	250.5 – 350.4	301 – 400
	350.5 – 500.4	401 – 500
	500.5 – 999.9	501 – 999
PM ₁₀	0 – 54	0 – 50
	55 – 154	51 – 100
	155 – 254	101 – 150
	255 – 354	151 – 200
	355 – 424	201 – 300
	425 – 504	301 – 400
	505 – 604	401 – 500
	605 – 999	501 – 999

Algorithm 2 Real-Time AQI Computation on the Sensor Node**Input:** Parsed PM_{2.5} and PM₁₀ concentrations**Output:** AQI value, descriptive category, display color*Per-Pollutant AQI Computation:*

- 1: **for** each pollutant in (PM_{2.5}, PM₁₀) **do**
- 2: Identify the concentration breakpoint containing the measured value
- 3: Compute AQI using Eq. (1)
- 4: **end for**

Final AQI Computation:

- 5: Set final AQI $\leftarrow \max(\text{AQI}_{PM_{2.5}}, \text{AQI}_{PM_{10}})$
- 6: Map final AQI to descriptive category (Table II)
- 7: Map final AQI to display color
- 8: Update TFT display
- 9: Broadcast AQI over mesh network
- 10: Return (final AQI, description, color)

To calculate AQI for each pollutant, the measured concentration is mapped to its corresponding index using linear interpolation between defined breakpoints

$$\text{AQI} = \frac{I_{\text{high}} - I_{\text{low}}}{C_{\text{high}} - C_{\text{low}}} \times (C - C_{\text{low}}) + I_{\text{low}} \quad (1)$$

where C_{low} and C_{high} are the lower and upper bounds of the pollutant concentration range, respectively, and I_{low} and I_{high} are the corresponding AQI values. The final AQI reported for each measurement is the higher value between PM_{2.5} and PM₁₀, reflecting the most critical air quality condition. Breakpoint values are summarized in Table III.

Algorithm 2 illustrates the real-time computation of AQI directly on the sensor node, detailing the step-by-step process by which PM measurements are converted into an AQI value, mapped to a descriptive category, and assigned a corresponding display color. Performing these computations locally allows the sensor node to provide immediate feedback while transmitting processed data over the network for further aggregation or analysis, ensuring low-latency reporting and reducing computational load on the central server.

III. METHODOLOGY

Modeling air pollution dynamics is challenging due to inherent complexity and nonlinearity. To address this, we adopt a data-driven Koopman operator framework. Using EDMD, dynamic states are estimated from time-series measurements collected by a spatially distributed PM sensor network. This setup captures dominant spatiotemporal patterns and provides comprehensive environmental coverage. By lifting nonlinear dynamics into a higher-dimensional linear space, Koopman EDMD offers a tractable, interpretable approximation of pollution evolution with minimal assumptions on the physical model.

Since EDMD alone does not handle real-time variability, measurement noise, or local disturbances, we integrate distributed Kalman filtering (DKF). Each sensor node performs recursive local estimation while collaborating to refine the global state. This hybrid approach combines Koopman-based predictive modeling with DKF robustness, yielding a reliable, scalable, and adaptive architecture for real-time air quality monitoring.

Note that Section III-A about the Koopman operator's theoretical framework presents the foundational concepts and notational conventions required for a clear and consistent understanding of the methodology and does not introduce any novel methodological or mathematical results [30], [36].

A. Koopman Operator Theory

Consider an uncontrolled nonlinear dynamical system evolving in the state space $\mathbb{R}^n$. Its evolution can be described in discrete-time form as

$$x_{k+1} = \mathcal{F}(x_k) \quad (2)$$

where $\mathcal{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ denotes the nonlinear transition function governing the system's dynamics. To facilitate the prediction of future states in such a nonlinear setting, the Koopman operator framework is employed. This approach enables a linear representation of the nonlinear dynamics by embedding the system into a higher-dimensional Hilbert space. The Koopman operator $\mathcal{K} : \mathcal{M} \rightarrow \mathcal{M}$ is formally defined as

$$(\mathcal{K}\psi)(x_k) = \psi(\mathcal{F}(x_k)) \quad (3)$$

where $\psi : \mathbb{R}^n \rightarrow \mathcal{M}$ is a set of observables or lifting functions that map the original state space into the Hilbert space $\mathcal{M}$.

The above formulation pertains to autonomous systems without external control inputs. To generalize the Koopman operator to controlled nonlinear dynamical systems, the state is augmented to incorporate control inputs. Let the extended state be defined as $\chi_k = [x_k^\top, u_k^\top]^\top$, where u_k represents the control input. The system dynamics are then reformulated as

$$\chi_{k+1} = \mathcal{H}(\chi_k) := \left[\mathcal{F}(x_k, u_k(0))^\top, \mathcal{S}u_k^\top \right]^\top. \quad (4)$$

Here, $\mathcal{S}$ is the left-shift operator acting on the sequence of control inputs, such that $(\mathcal{S}u)_k = u_{k+1}$, where u_k denotes the k th element in the input sequence. This extension allows the Koopman operator framework to be applied in the presence of control, enabling linear analysis and prediction of controlled nonlinear systems.

The Koopman operator for the extended system dynamics can be redefined as

$$(\mathcal{K}\Theta)(\chi) = \Theta(\mathcal{H}(\chi)) \quad (5)$$

where $\Theta : \mathbb{R}^{n+\ell} \rightarrow \mathbb{R}$ is a transfer function (observable) that maps the extended state $\chi \in \mathbb{R}^{n+\ell}$, consisting of both the system state and control input, into a suitable Hilbert space in which the Koopman operator $\mathcal{K} : \mathcal{M} \rightarrow \mathcal{M}$ operates linearly. This formulation captures the dynamics of a controlled nonlinear system in a linear framework.

B. Extended Dynamic Mode Decomposition

The precise mathematical representation of air pollution dynamics is commonly expressed in the state-space form as

$$\chi_{k+1} = \begin{bmatrix} f(x_k, u_k) + w_k \\ Su_k \end{bmatrix}, \quad y_k = h(x_k) + v_k \quad (6)$$

where $\chi_k = [x_k^\top, u_k^\top]^\top \in \mathbb{R}^n$ denotes the system state, $u_k \in \mathbb{R}^m$ represents external inputs such as wind speed and humidity, and w_k and v_k are process and measurement noise, respectively. The function $f(\cdot)$ captures the underlying system dynamics, including pollutant deposition, atmospheric transport, and source emission rates, while $h(\cdot)$ is a nonlinear mapping from the state space to observed pollution levels. In practice, accurately characterizing these dynamics is challenging due to their inherent nonlinearity and the presence of unknown or time-varying environmental factors such as wind direction and humidity. To address these limitations and enable predictive modeling of the air pollution process, we utilize the EDMD algorithm to approximate the Koopman operator $\mathcal{K}$. Specifically, EDMD estimates $\mathcal{K}$ in the following form:

$$\Theta(\chi_{k+1}) = \mathcal{K}^\top \Theta(\chi_k) + \epsilon(\chi_k) \quad (7)$$

where $\Theta : \mathbb{R}^{n+\ell} \rightarrow \mathbb{R}$ is a transfer function previously defined and $\epsilon(\chi_k)$ is the approximation error of the Koopman operator. To approximate the Koopman operator $\mathcal{K}$, we seek a matrix representation $K \in \mathbb{R}^{N_\Theta \times N_\Theta}$ that minimizes the discrepancy between predicted and observed lifted states. Specifically, we formulate the following least-squares objective:

$$E_K = \sum_{k=1}^{N_K} \|\epsilon(\chi_k)\|^2 = \sum_{k=1}^{N_K} \left\| \Theta(\chi_{k+1}) - K^\top \Theta(\chi_k) \right\|^2 \quad (8)$$

where $\Theta(\chi_k) = [\theta_1(\chi_k), \dots, \theta_{N_\Theta}(\chi_k)]^\top \in \mathbb{R}^{N_\Theta}$ is a vector of basis functions that upgrade the original system state $\chi_j \in \mathbb{R}^{n+\ell}$ to a higher-dimensional feature space in which the dynamics appear linear.

Given a collection of data snapshots from the system, we define

$$\begin{aligned} X_K &= [\Theta(\chi_1), \dots, \Theta(\chi_{N_K-1})] \in \mathbb{R}^{N_\Theta \times (N_K-1)} \\ Y_K &= [\Theta(\chi_2), \dots, \Theta(\chi_{N_K})] \in \mathbb{R}^{N_\Theta \times (N_K-1)} \\ U_K &= [\Theta(u_1), \dots, \Theta(u_{N_K-1})] \in \mathbb{R}^{N_\Theta \times (N_K-1)}. \end{aligned}$$

Here, $\chi_{k+1} = \mathcal{H}(\chi_k, u_k)$ denotes the extended system dynamics that couples state evolution with the input.

To identify a linear approximation of the system in the lifted space, we solve for the matrices $A \in \mathbb{R}^{N_\Theta \times N_\Theta}$ and

$B \in \mathbb{R}^{N_\Theta \times N_\Theta}$ by minimizing the error in (8). The closed-form solution is given by

$$[A, B] = Y_K \begin{bmatrix} X_K \\ U_K \end{bmatrix}^\dagger \quad (9)$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. The resulting matrices A and B serve as approximations of the lifted system's drift and input dynamics, respectively, within the Koopman-based linear framework. After lifting the extended state $\chi_k = [x_k^\top, u_k^\top]^\top \in \mathbb{R}^{n+m}$ into a higher-dimensional feature space using a set of basis functions (observables) $\Theta(\cdot)$, the nonlinear air pollution dynamics can be approximated by a discrete-time linear system in the observable space

$$z_{k+1} = Az_k + B\tilde{u}_k + w_{z,k}, \quad y_k = Cz_k + v_k \quad (10)$$

where $z_k = \Theta(\chi_k) \in \mathbb{R}^{N_\Theta}$ is the lifted state (observable by Kukowski), $\tilde{u}_k = \Theta(u_k) \in \mathbb{R}^{N_u}$ is a lifted representation of the input u_k , $y_k \in \mathbb{R}^p$ is the measured output, $C = \chi_k X_K^\dagger \in \mathbb{R}^{p \times N_\Theta}$ is the output matrix, $w_{z,k} \sim \mathcal{N}(0, Q_z)$ is the process noise in the lifted space, and $v_{z,k} \sim \mathcal{N}(0, R)$ is the measurement noise. We assume that both noises are zero-mean white Gaussian noise (WGN), uncorrelated with each other.

Remark 2: In the context of air pollution dynamics, the term $\tilde{u}_k$ in the Koopman linearized model typically represents external inputs such as wind speed, humidity, or temperature. Unlike classical control systems, where $\tilde{u}_k$ denotes an actuated control input, these environmental variables are not controllable but rather exogenous disturbances driven by natural processes. Therefore, interpreting $\tilde{u}_k$ as a control signal is not appropriate in this setting. If the effect of such environmental factors is negligible, slowly varying, or uncorrelated with the system state over the prediction horizon, the Koopman model reduces to an autonomous form $z_{k+1} = Az_k + w_{z,k}$ in which the lifted state evolution is governed solely by the internal dynamics. This simplification enables state estimation using a standard Kalman filter without requiring measurements or modeling of $\tilde{u}_k$.

C. KDKF Architecture

Although the nonlinear dynamics of the air pollution system are approximated using the Koopman operator framework, the presence of process disturbances and measurement noise, as indicated in (9), remains a significant challenge in practice. These uncertainties can degrade the quality of state predictions when relying solely on the model. To address this, the Kalman filter provides an effective solution by optimally combining model-based predictions with noisy sensor measurements to produce refined state estimates. Moreover, given the distributed deployment of sensor motes and their inherent communication capabilities, it is advantageous to exploit their interconnectivity. A distributed Kalman filter (DKF) can leverage local observations at each mote while also incorporating information exchanged with neighboring motes, leading to improved global estimation accuracy. This distributed framework not only enhances robustness to sensor-level noise and model mismatch but also supports scalability and resilience in large-scale air quality monitoring networks.

D. AI-Augmented Koopman–Kalman Filter (AI-KF)

Air quality monitoring is challenging because pollutant dynamics are nonlinear, stochastic, and influenced by environmental factors such as temperature, humidity, and wind. Traditional estimation techniques like the classical Kalman filter (KF) perform well for linear systems with Gaussian noise but degrade under nonlinear dispersion, sensor drift, and dynamically changing real-world conditions.

To overcome these limitations, we propose an *AI-augmented Koopman–Kalman filter* (AI-KF) framework that integrates three complementary layers: 1) a Koopman lifting stage that embeds nonlinear pollutant dynamics into a higher-dimensional linear space; 2) a distributed Koopman–Kalman filter for consensus-based state estimation across multiple sensing nodes; and 3) a neural residual corrector that learns and compensates for systematic modeling errors left unresolved by the Koopman-KF.

1) Koopman Lifting of Nonlinear Dynamics: Consider the air quality system governed by hidden nonlinear dynamics

$$x_{k+1} = f(x_k, u_k) + w_k, \quad y_k = h(x_k) + v_k \quad (11)$$

where $x_k \in \mathbb{R}^n$ is the state vector (e.g., PM2.5 and PM10 concentrations), $u_k \in \mathbb{R}^m$ represents exogenous inputs (e.g., CO₂, temperature, and humidity), $y_k \in \mathbb{R}^p$ are sensor measurements, and w_k, v_k denote process and measurement noise, respectively.

Instead of directly estimating nonlinear dynamics, the Koopman operator theory introduces a lifting function $\psi : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^d$ that maps original states and inputs into an expanded observable space

$$z_k = \psi(x_k, u_k) = \left[x_k^\top, u_k^\top, x_k^{\odot 2}, u_k^{\odot 2}, \text{kron}(x_k, u_k) \right]^\top \quad (12)$$

where $d \gg n$, and the notation $\odot$ denotes element-wise exponentiation.

In this lifted space, nonlinear dynamics can be approximated by a linear operator K

$$z_{k+1} \approx K z_k. \quad (13)$$

This transformation allows us to employ linear estimation methods, such as the Kalman filter, on otherwise nonlinear pollutant dynamics.

2) Distributed Koopman–Kalman Filter: Given N spatially distributed sensor nodes monitoring air quality, each node maintains a local estimate of the lifted state z_k . The standard Kalman filter prediction-update cycle in the lifted domain is

$$\text{Prediction:} \quad \hat{z}_{k|k-1} = A \hat{z}_{k-1|k-1} \quad (14)$$

$$P_{k|k-1} = A P_{k-1|k-1} A^\top + Q_z \quad (15)$$

$$\text{Update:} \quad K_k = P_{k|k-1} C^\top (C P_{k|k-1} C^\top + R)^{-1} \quad (16)$$

$$\hat{z}_{k|k} = \hat{z}_{k|k-1} + K_k (y_k - C \hat{z}_{k|k-1}) \quad (17)$$

$$P_{k|k} = (I - K_k C) P_{k|k-1} \quad (18)$$

where $A \in \mathbb{R}^{d \times d}$ and $C \in \mathbb{R}^{p \times d}$ are the estimated Koopman system matrices and Q_z and R are process and measurement noise covariances, respectively.

To ensure consistency across nodes, we implement a *consensus averaging step*

$$\hat{z}_{k|k}^{(i)} \leftarrow \frac{1}{N} \sum_{j=1}^N \hat{z}_{k|k}^{(j)}, \quad i = 1, \dots, N \quad (19)$$

enforcing cooperation between sensing units in the network.

3) Neural Residual Corrector: While the Koopman-KF linearizes nonlinearities effectively, mismatches remain due to unmodeled dynamics and noise. To address this, we introduce a neural residual learning module that corrects the systematic error

$$r_k = x_k - \hat{x}_{k|k} \quad (20)$$

where $\hat{x}_{k|k} = C \hat{z}_{k|k}$ is the reconstructed state.

A lightweight feedforward NN is trained online (or periodically offline) to approximate $r_k = \mathcal{N}(z_k)$. The corrected AI-KF estimate is then

$$\hat{x}_k^{\text{AI-KF}} = \hat{x}_{k|k} + \mathcal{N}(z_k). \quad (21)$$

To implement the neural residual learning component, a compact feed-forward architecture with two hidden layers (32 and 16 neurons with ReLU activation, respectively) was employed. Training was performed offline using the indoor air-quality dataset, divided into 70% training, 15% validation, and 15% testing partitions. The network was optimized using the Adam optimizer with an initial learning rate of 10^{-3} and a batch size of 64. The mean squared error (mse) between the true state and the KF estimate was used as the loss function, and early stopping was applied to prevent overfitting. Once trained, the network operates in real time through forward propagation only, introducing minimal computational overhead while providing nonlinear correction capability. The detailed algorithm is provided in Algorithm 3.

The inclusion of the residual neural corrector enables the estimator to learn unmodeled behavior from data while preserving the advantages of the physically interpretable Koopman-KF structure.

The proposed AI-KF combines model-based estimation with machine learning. The Koopman operator embeds nonlinear pollutant dynamics into a linear domain, preserving Kalman filter structure and interpretability. Its distributed implementation ensures scalability for sensor networks in smart cities or indoor monitoring. The neural residual corrector offers adaptive learning, addressing uncertainties and disturbances while maintaining real-time performance.

This balance of interpretability, scalability, and adaptability makes AI-KF a powerful tool for next-generation environmental sensing.

E. Robust Estimation With Missing Measurements

To ensure robust estimation under incomplete or faulty measurements, the proposed AI-augmented Koopman–Kalman filter explicitly incorporates a missing-data handling mechanism. At each time step, measurement validity is assessed and encoded through a binary flag v_k , with a corresponding missing-data counter m_k that evolves as $m_k = m_{k-1} + 1$ when measurements are invalid and resets to zero otherwise. State

Algorithm 3 AI-Augmented Koopman–Kalman Filter (AI-KF) With Missing Data Handling

Input: Sensor data $\{y_k, u_k\}_{k=1}^T$, Koopman matrices (A, C) , noise covariances (Q_z, R) , number of nodes N , missing data threshold τ

Output: Enhanced state estimates $\{\hat{x}_k^{\text{AI-KF}}\}$, measurement validity flags $\{v_k\}$

Initialization:

- 1: Lifted state: $z_0 = 0$ for each node i
- 2: Covariance: $P_0 = I$ for each node i
- 3: Residual neural network: $\mathcal{N}(\cdot)$
- 4: Consecutive missing counter: $m \leftarrow 0$

Time-Step Loop:

- 5: **for** $k = 1$ to T **do**
 - 6: **Measurement Validation:**
 - 7: **if** y_k is NaN or $\|y_k - C\hat{z}_{k-1|k-1}\| > \tau$ **then**
 - 8: $v_k \leftarrow 0$ {Invalid measurement flag}
 - 9: $m \leftarrow m + 1$
 - 10: **else**
 - 11: $v_k \leftarrow 1$ {Valid measurement flag}
 - 12: $m \leftarrow 0$
 - 13: **end if**
 - 14: **for** each node $i = 1, \dots, N$ **do**
 - 15: **Prediction:**
 - 16: $\hat{z}_{k|k-1}^{(i)} = A\hat{z}_{k-1|k-1}^{(i)}$
 - 17: $P_{k|k-1}^{(i)} = AP_{k-1|k-1}^{(i)}A^\top + Q_z$
 - 18: **Adaptive Update:**
 - 19: **if** $v_k = 1$ **then** {Normal update with measurement}
 - 20: $K_k^{(i)} = P_{k|k-1}^{(i)}C^\top(CP_{k|k-1}^{(i)}C^\top + R)^{-1}$
 - 21: $\hat{z}_{k|k}^{(i)} = \hat{z}_{k|k-1}^{(i)} + K_k^{(i)}(y_k - C\hat{z}_{k|k-1}^{(i)})$
 - 22: $P_{k|k}^{(i)} = (I - K_k^{(i)}C)P_{k|k-1}^{(i)}$
 - 23: **else** {Prediction-only mode for missing data}
 - 24: $\hat{z}_{k|k}^{(i)} = \hat{z}_{k|k-1}^{(i)}$
 - 25: $P_{k|k}^{(i)} = P_{k|k-1}^{(i)} \times (1 + \alpha \cdot \min(m, M))$ {Uncertainty inflation}
 - 26: **end if**
 - 27: **end for**
 - 28: **Consensus Averaging:**
 - 29: $\hat{z}_{k|k}^{(i)} \leftarrow \frac{1}{N} \sum_{j=1}^N \hat{z}_{k|k}^{(j)}$ for all i
 - 30: **Reconstruction:**
 - 31: $\hat{x}_{k|k} = C\hat{z}_{k|k}$
 - 32: **Adaptive Residual Correction:**
 - 33: $\beta = \exp(-\lambda \cdot m)$ {Confidence decay for extended gaps}
 - 34: $\hat{x}_k^{\text{AI-KF}} = \hat{x}_{k|k} + \beta \cdot \mathcal{N}(\hat{z}_{k|k})$
 - 35: **Uncertainty Propagation:**
 - 36: **if** $m > 0$ **then** report elevated uncertainty bound:
 - 37: $\sigma_k = \sqrt{\text{tr}(P_{k|k})}$
 - 37: **end for**
-

prediction for each node is performed in the lifted Koopman space according to

$$\hat{z}_{k|k-1}^{(i)} = A\hat{z}_{k-1|k-1}^{(i)}, \quad P_{k|k-1}^{(i)} = AP_{k-1|k-1}^{(i)}A^\top + Q_z \quad (22)$$

followed by a conditional update that applies the standard Kalman correction when $v_k = 1$ and reverts to prediction-only propagation when $v_k = 0$. During measurement gaps, estimator overconfidence is prevented by adaptive covariance inflation

$$P_{k|k}^{(i)} = P_{k|k-1}^{(i)}(1 + \alpha \min(m_k, M)) \quad (23)$$

while the state estimate remains unchanged. Local lifted-state estimates are then fused via consensus averaging, and the physical state is reconstructed as $\hat{x}_{k|k} = C\hat{z}_{k|k}$. To compensate for unmodeled nonlinearities, a residual neural correction is applied at the reconstruction stage

$$\hat{x}_k^{\text{AI-KF}} = \hat{x}_{k|k} + \beta_k \mathcal{N}(\hat{z}_{k|k}), \quad \beta_k = \exp(-\lambda m_k) \quad (24)$$

where the exponentially decaying confidence factor β_k attenuates the neural contribution during extended measurement outages. This design preserves the linear Koopman–Kalman structure, supports distributed consensus, and ensures graceful performance degradation under intermittent sensing.

Remark 3: While contemporary ensemble-based methods such as FedEI [7] have demonstrated strong performance in reconstructing extended missing data intervals through iterative model selection and aggregation, they are primarily tailored to offline or batch-oriented reconstruction settings. In contrast, the proposed approach is explicitly designed for real-time estimation, providing causal and robust state updates at each time step despite intermittent sensor failures, without requiring access to future measurements or retrospective reprocessing.

IV. SENSOR DEPLOYMENT SITE AND DATA COLLECTION

A. Sensor Deployment Site

The sensor deployment site significantly influences data representativeness. Sensors were deployed in a controlled indoor environment (4.7×4.2 m), simulating conditions where vulnerable populations may be exposed to PM. This setup allows testing under realistic indoor airflow, moderate human activity, and controlled pollutant sources such as dust or smoke.

Sensor placement, shown in Fig. 4, ensures spatial coverage to capture local PM variations and assess network performance in a confined space. Future deployments could include open-floor areas such as emergency rooms, laboratories handling chemicals or nanoelectronics, and other sensitive environments requiring localized AQI monitoring and early warnings.

B. Data Collection

Data were collected over multiple sessions on three dates, each lasting at least 2 h, with sensor nodes recording measurements every 5 s. Controlled events included dust emissions, simulated smoke, and routine indoor activity to validate data parsing, AQI computation, and mesh network broadcasting. Data were logged locally and analyzed in Python.

Each dataset includes PM1.0, PM2.5, PM10, and computed AQI values, forming the basis for statistical analysis, correlation studies, and visualization after KDKF and AI-enhanced

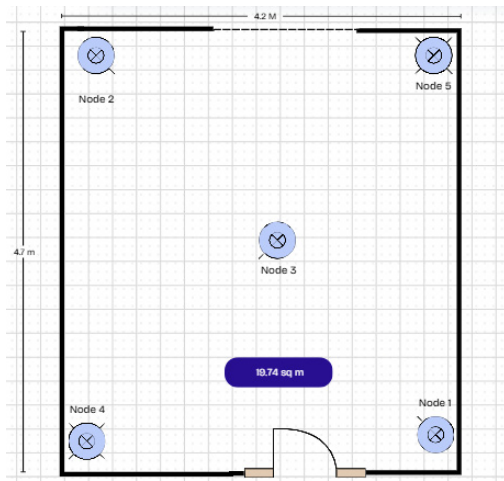


Fig. 4. Floor plan of the deployment site (4.7×4.2 m) showing sensor node placement for AQI monitoring.

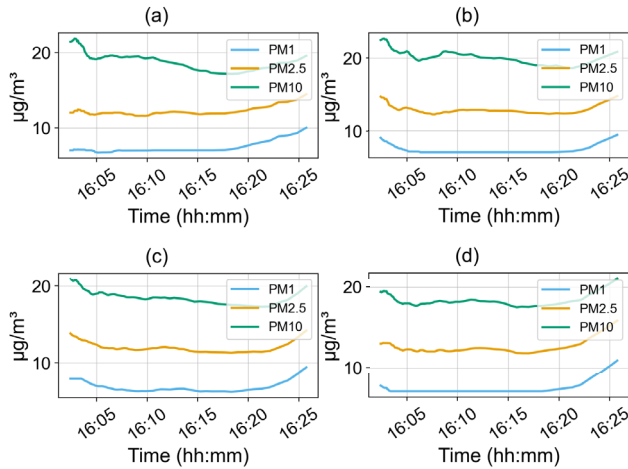


Fig. 5. Temporal PM measurements from four sensor nodes over a 20-min interval on July 9, 2025, after 16:00. (a) Node 1. (b) Node 2. (c) Node 4. (d) Node 5.

DKF implementation. Representative visualizations illustrate temporal deviations across nodes. A 30-min sample under normal indoor conditions is shown in Fig. 5. Daily maximum AQI values per node are in Fig. 6, and spatial AQI distributions across nodes during one session are shown in Fig. 7.

We extended the analysis to the full system with all five nodes operating simultaneously on two dates, September 21 and September 23, over several hours. These sessions captured varying conditions, including temperature and humidity changes, as well as pollutant introduction, providing a dataset to evaluate the performance of the different Kalman filter approaches under diverse exposure scenarios.

V. RESULTS FOR AIR QUALITY ASSESSMENT

A. Performance Evaluation of AI-KF

The effectiveness of the AI-augmented Koopman–Kalman filter (AI-KF) was evaluated against the baseline Koopman-KF using quantitative error metrics and qualitative visualizations, highlighting both numerical gains and improved practical reliability.

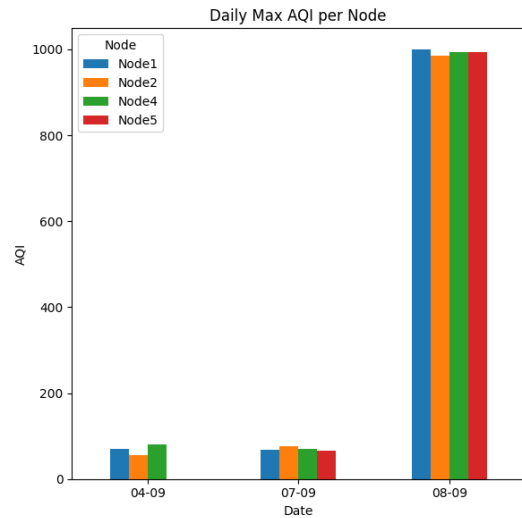


Fig. 6. Daily maximum AQI values recorded by individual sensor nodes on separate dates.

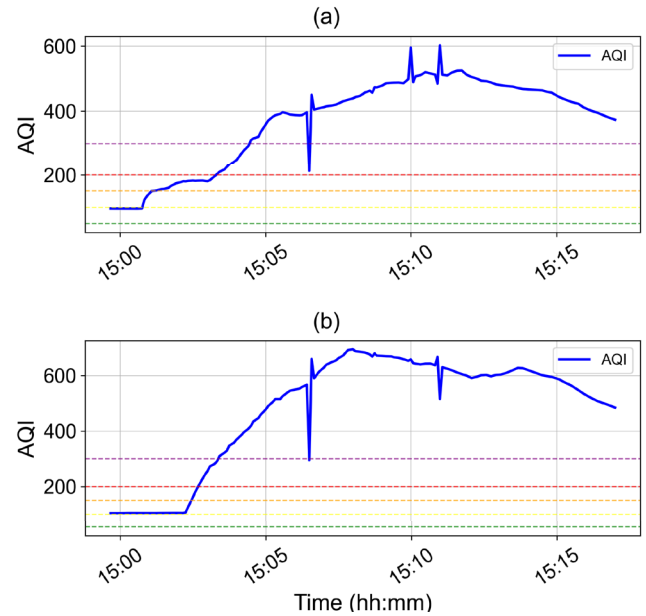


Fig. 7. AQI results from different sensor nodes, each showing their corresponding air quality categories. (a) Node 4 on August 9, 2025. (b) Node 5 on September 8, 2025.

RMSE values for both methods across two representative dates are shown in Table IV. AI-KF consistently outperforms Koopman-KF: on September 21 (13:20–16:00), PM_{2.5} and PM₁₀ RMSEs drop from 51.243 and 50.504 $\mu\text{g}/\text{m}^3$ to 3.3648 and 3.3927 $\mu\text{g}/\text{m}^3$, respectively; on September 23 (17:12–19:23), they decrease from 3.512 and 3.878 to 1.197 and 1.988 $\mu\text{g}/\text{m}^3$, demonstrating effective compensation for nonlinearities and model mismatches.

Fig. 8 illustrates that AI-KF estimates (dashed-dotted red) track ground truth (solid black) more closely than Koopman-KF (dashed blue), particularly during rapid concentration changes.

Table V reports additional metrics including NRMSE, percent RMSE, MAPE, and relative improvement. For Date A,

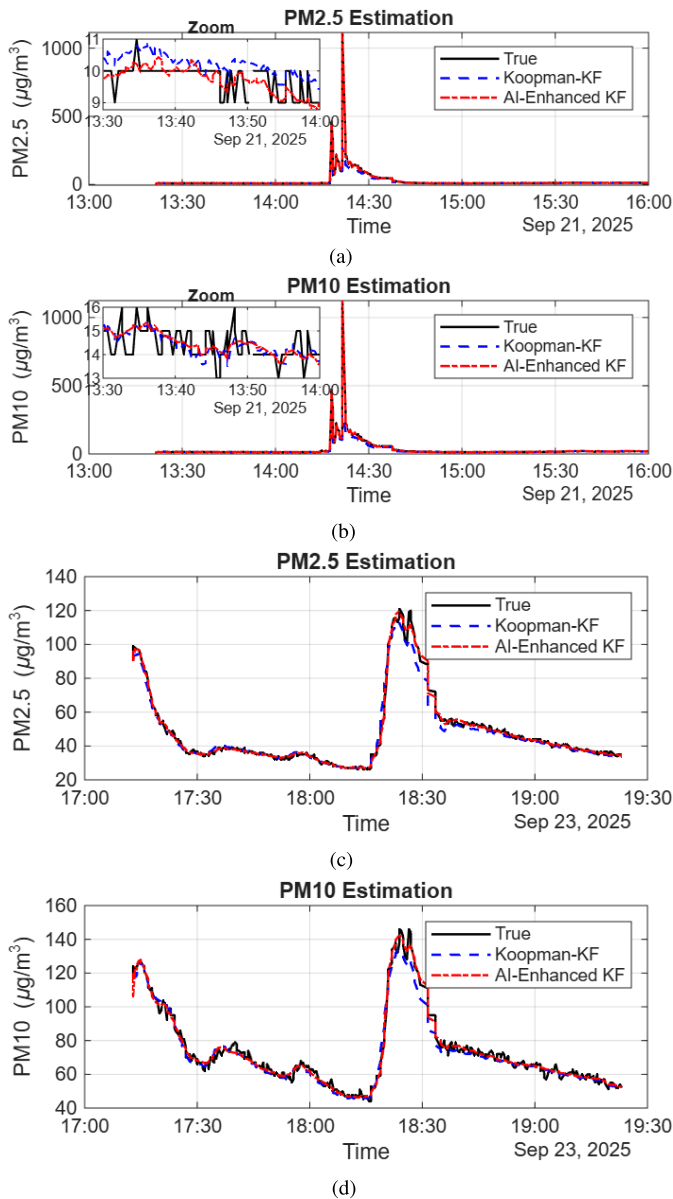


Fig. 8. Comparison of true states, Koopman-KF estimates, and AI-KF enhanced estimates for PM2.5 and PM10 at two different datetimes. (a) PM2.5—Date A (September 21, 13:20–16:00). (b) PM10—Date A (September 21, 13:20–16:00). (c) PM2.5—Date B (September 23, 17:12–19:23). (d) PM10—Date B (September 23, 17:12–19:23).

AI-KF reduces the RMSE by over 93% for both pollutants. The NRMSE decreases from 0.046331 (PM2.5) and 0.045458 (PM10) under Koopman-KF to 0.0030423 and 0.0030537, respectively, under AI-KF. Percent RMSE drops from 190.2% to 12.489% (PM2.5) and from 155.61% to 10.453% (PM10), confirming improvements are genuine, not scale-dependent. MAPE also decreases significantly, despite sensitivity to near-zero values.

Qualitative insights from the figures include the following.

- 1) *Time Series (Fig. 8)*: AI-KF tracks true dynamics more closely, especially during rapid changes.
- 2) *Per-Node RMSE (Fig. 9)*: Node-level variability is suppressed by consensus and AI correction.
- 3) *Summary RMSE (Fig. 10)*: Overall error reduction clearly demonstrates AI-KF superiority.

TABLE IV

RMSE VALUES OF KOOPMAN-KF AND AI-KF APPROACHES FOR PM2.5 AND PM10 DURING SELECTED HOURS ON TWO DIFFERENT DATES. (a) DATE A (SEPTEMBER 21, 13:20–16:00). (b) DATE B (SEPTEMBER 23, 17:12–19:23)

(a)

Method	PM2.5	PM10
Koopman-KF	51.243	50.504
AI-KF	3.3648	3.3927

(b)

Method	PM2.5	PM10
Koopman-KF	3.512	3.878
AI-KF	1.197	1.988

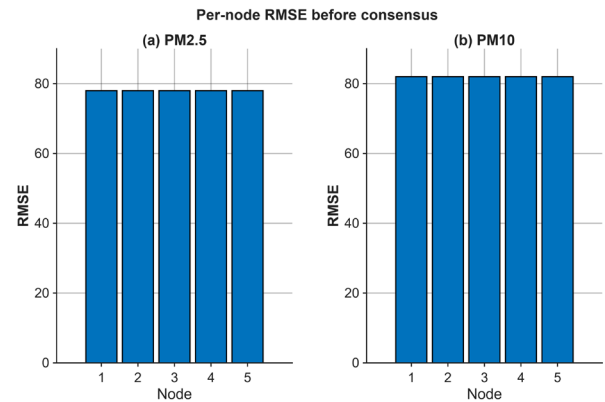


Fig. 9. Per-node RMSE distribution before consensus. (a) PM2.5. (b) PM10.

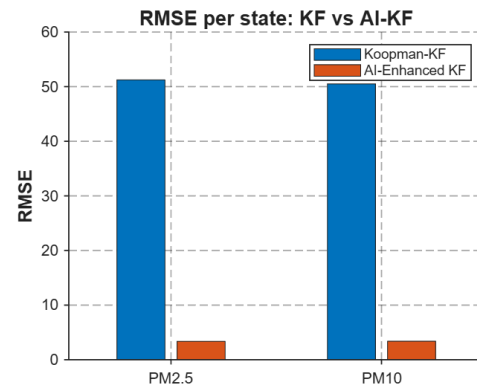


Fig. 10. Summary comparison of RMSE (Koopman-KF versus AI-KDKF).

- 4) *Residuals (Fig. 11)*: Bias and variance are reduced, with residuals approaching white noise.
- 5) *Predicted Versus Measured Scatter (Fig. 12)*: AI-KF aligns more tightly along the identity line with higher R^2 .

Overall, the results indicate that integrating residual learning with Koopman-based distributed state estimation yields substantial improvements. Error metrics are reduced by over 93%, and internode consistency is enhanced, demonstrating the AI-KDKF framework's effectiveness for reliable and scalable IAQ monitoring. These gains are particularly important for real-world deployments, where both accuracy and robustness are essential for health-related decision-making.

TABLE V
COMPREHENSIVE ERROR METRICS FOR KOOPMAN-KF AND AI-KF ON DATE A (SEPTEMBER 21, 13:20–16:00)

State	RMSE_KF	RMSE_AI	NRMSE_R_KF	NRMSE_R_AI	%RMSE_KF	%RMSE_AI	MAPE_KF	MAPE_AI	Improv.%
PM2.5	51.243	3.3648	0.046331	0.0030423	190.2	12.489	8.0432	3.909	93.434
PM10	50.504	3.3927	0.045458	0.0030537	155.61	10.453	6.8982	5.0188	93.282

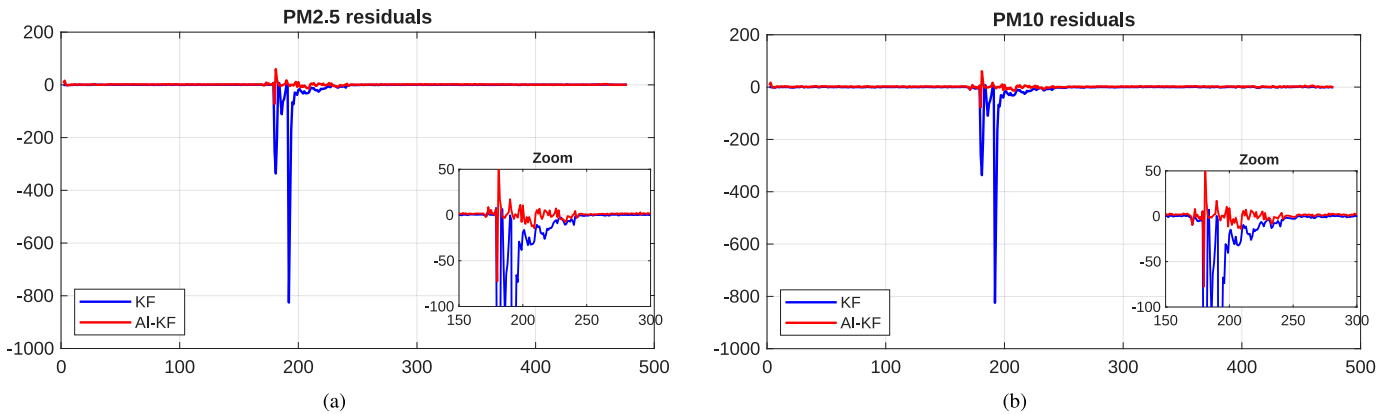


Fig. 11. Residual error analysis for Koopman-KF versus AI-KDF. (a) PM2.5. (b) PM10.

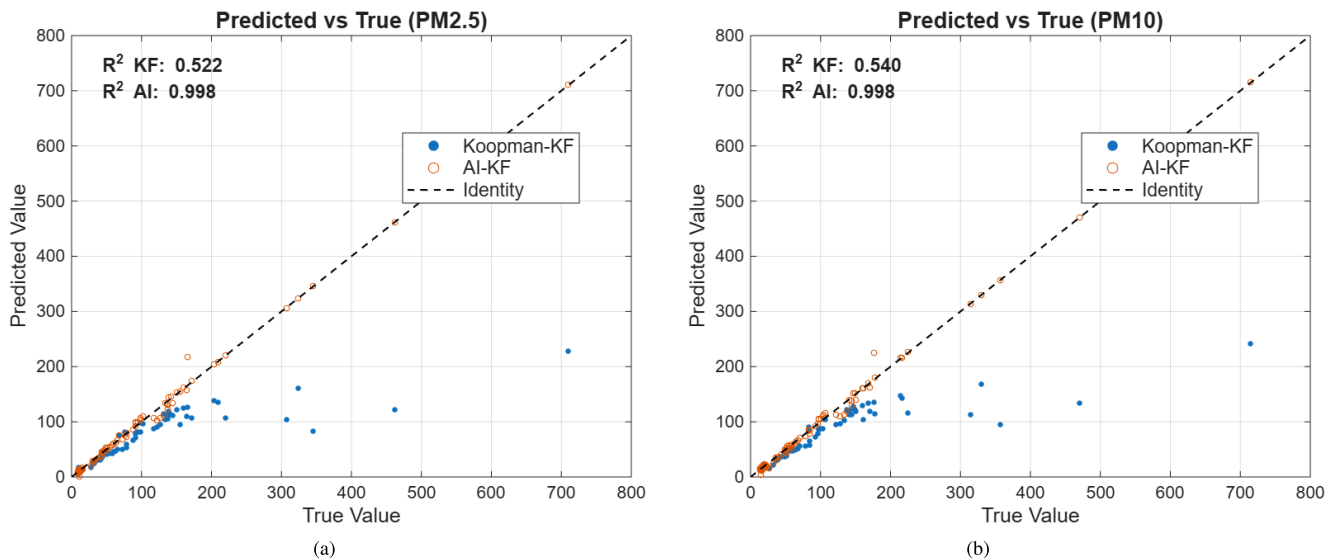


Fig. 12. Predicted versus measured scatter plots for both pollutants. (a) PM2.5. (b) PM10.

In addition to numerical accuracy, we evaluated environmental relevance using the AQI, which integrates PM2.5 and PM10 measurements into a standardized metric. For Date A, four time points captured a peak event from a localized smoking experiment: one before, two during, and one after the peak. Fig. 13 compares the AQI values obtained from ground-truth measurements, Koopman-KF, and AI-KF.

The AI-KF closely tracks true AQI peaks and remains consistent with PM measurements, accurately reflecting hazardous events. In contrast, the Koopman-KF underestimates peaks and exhibits larger deviations, reducing environmental fidelity. These results confirm that AI-based residual correction enhances both numerical accuracy and practical suitability for real-world air quality monitoring.

B. Discussion

Our air quality WSN consists of five PM3006S sensor nodes arranged in a mesh network using ESP TTGO T-Display

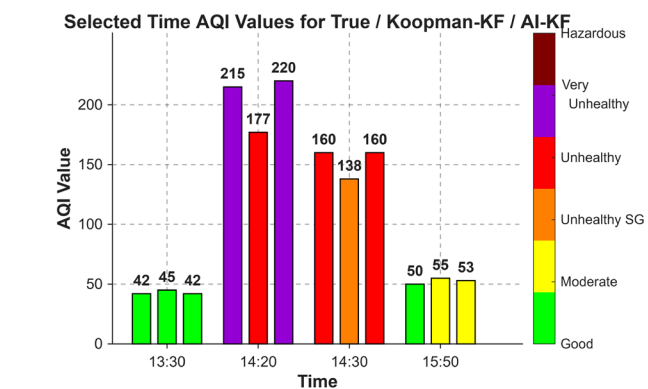


Fig. 13. AQI values at selected times (true, Koopman-KF, AI-KF) on Date A.

microcontrollers. A root node aggregates measurements and forwards them to a local router, where data are logged per

node. Due to internode variability, a distributed Kalman filter (DKF) is employed to obtain a unified estimate, enhanced with the Koopman operator to address system nonlinearity and further refined using an AI-based residual correction.

Two experimental scenarios are evaluated: isolated node testing to assess stability under minor disturbances, and a full multinode deployment under varying environmental conditions. While the Koopman-DKF captures nonlinear dynamics effectively, residual errors persist and are compensated by a lightweight NN.

Quantitatively, the AI-enhanced DKF yields smoother and more robust estimates, reducing RMSE by approximately 93% in the first experiment and achieving $3\times$ and $2\times$ improvements for PM_{2.5} and PM₁₀, respectively, in the second. These gains also translate to AQI estimation, where peak events are tracked more faithfully, confirming the framework's robustness, accuracy, and practical relevance.

While other works on distributed Kalman filtering and alternative estimation methodologies, such as [13] and [27], focused primarily on extended-fractional KF applications, they often required detailed information about transfer function coefficients and well-characterized chemical pollutants. In contrast, our system relies on simplified PM-only measurements at various scales. The Koopman-DKF provides a solution to overcome the limited information for estimation, and while Koopman EDMD offers a valuable model approximation, it does not inherently account for real-time variability, measurement noise, or local disturbances. To address these limitations, we integrate a DKF scheme into the framework, allowing each sensor node to perform recursive local estimation while collaboratively improving global state accuracy.

The EKF is particularly suitable for accurate state estimation in highly nonlinear systems, where additive uncertainties, initial deviations, noise, disturbances, and missing measurements affect prediction performance. However, applications of AI-enhanced Kalman filters in sensor fusion remain limited, with only a few trials highlighting the advantages of integrating AI for residual correction and real-time improvement [19], [35].

Our results show that integrating AI into the Koopman-DKF framework improves WSN performance and real-time accuracy over AI-free implementations. By reducing estimation errors, smoothing transient fluctuations, and better capturing pollutant peaks, the AI-enhanced approach provides more accurate PM_{2.5}/PM₁₀ and AQI estimates, supporting reliable monitoring and health-related decision-making.

It is important to point out, however, that the accuracy of the optical PM sensors is fundamentally challenged by dynamic atmospheric processes that alter particle properties. Specifically, coagulation and condensation continuously modify the particle size distribution. Since the sensor's laser scattering signal is highly dependent on particle size, these changes introduce significant, nonlinear errors in the reported mass concentration. Furthermore, deposition can cause systematic underestimation, particularly for larger PM₁₀ particles [17], [23]. Explicitly modeling these phenomena within a traditional Kalman filter framework is prohibitively complex, as it would require tracking multiple particle size bins and

solving computationally expensive equations in real-time. Our AI-augmented Kalman filter addresses this fundamental limitation through a hybrid structure: the core Kalman filter provides a robust, model-based estimate of PM concentration using measurable inputs like humidity, while a co-processing NN dynamically learns and compensates for the complex errors induced by the unmodeled physicochemical processes. This approach allows the system to implicitly capture the net effect of coagulation, condensation, and deposition on the sensor's signal, effectively translating the raw, biased optical data into a physically accurate measurement without the need for an intractably complex analytical model.

VI. CONCLUSION

This study demonstrates that integrating AI into the Koopman-based distributed Kalman filter (DKF) significantly enhances the performance of IAQ monitoring systems. By combining physics-informed modeling with a lightweight NN residual corrector, the AI-enhanced DKF (AI-KDKF) effectively addresses nonlinearities, sensor noise, and local disturbances that are commonly present in real-world environments.

Experimental results on a five-node WSN indicate that the AI-KDKF consistently reduces estimation errors for PM_{2.5} and PM₁₀ compared to the standard Koopman-KF. RMSE reductions of over 93% were observed, alongside improvements in normalized errors and mean absolute percentage errors, confirming both numerical and practical gains. The AI-based correction enables the filter to capture transient pollutant peaks more accurately, leading to AQI estimations that closely reflect real environmental conditions.

Beyond numerical accuracy, the proposed approach demonstrates clear advantages for actionable environmental monitoring. By providing smoother, more reliable, and timely estimates of particulate concentrations, the AI-KDKF supports better-informed decision-making for IAQ management, health risk assessment, and regulatory compliance. Importantly, the method retains the interpretability and modularity of the underlying Kalman framework while enhancing adaptability through data-driven intelligence.

In summary, the integration of AI with Koopman-DKF offers a scalable, robust, and accurate solution for real-time IAQ assessment. This hybrid approach represents a promising avenue for future research in intelligent environmental sensing, sensor fusion, and smart building applications, where both precision and resilience are critical.

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